

* Recap: $\vec{r}(t) = (x(t), y(t), z(t))$ (position - vector)

• Velocity: $\frac{d\vec{r}}{dt} = \vec{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$

acceleration: $\frac{d^2\vec{r}}{dt^2} = \frac{d\vec{v}}{dt} = \vec{a}(t) = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}, \frac{d^2z}{dt^2} \right)$

$$\vec{r}(t) = (x(t), y(t), z(t)) = (x(t), 0, 0) + (0, y(t), 0) + (0, 0, z(t))$$

$$= x(t) \underbrace{(1, 0, 0)}_{\hat{e}_x} + y(t) \underbrace{(0, 1, 0)}_{\hat{e}_y} + z(t) \underbrace{(0, 0, 1)}_{\hat{e}_z}$$

$$\vec{F} = m\vec{a} = m \frac{d^2\vec{r}}{dt^2} = x(t)\hat{e}_x + y(t)\hat{e}_y + z(t)\hat{e}_z$$

some force: depends on problem

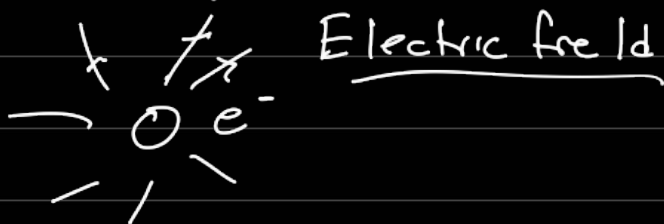
→ Lorentz - force: Force felt on a charged particle moving in a uniform magnetic field

$$\vec{F}_{\text{mag}} = q \vec{v} \times \vec{B}$$

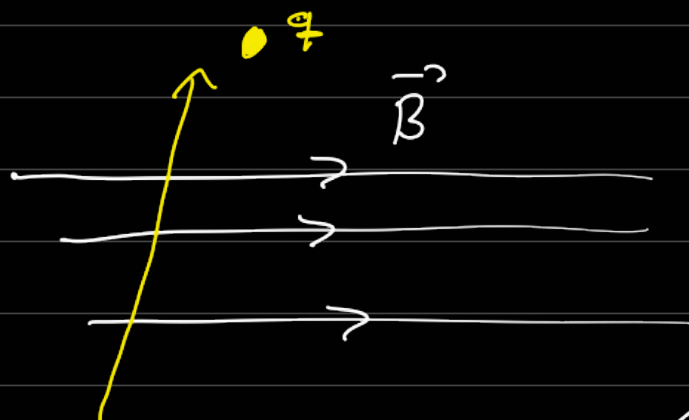
q : charge of particle

$\vec{v}$: Velocity of the same particle

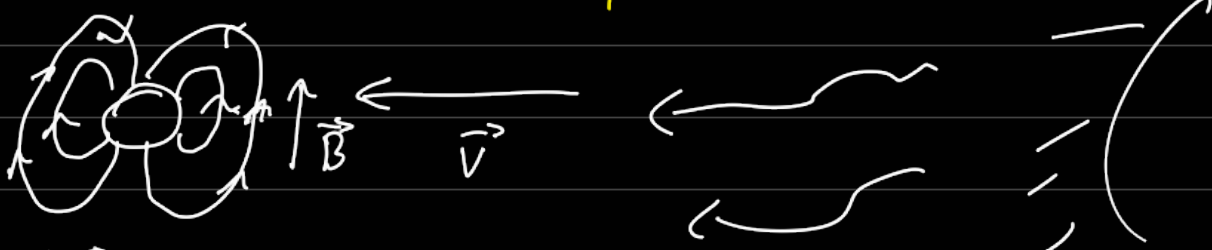
$\vec{B}$: Magnetic field vector



Magnetic field:



• Ex: "Northern lights"



$$m \frac{d^2\vec{r}}{dt^2} = \vec{F} = q \vec{v} \times \vec{B}$$

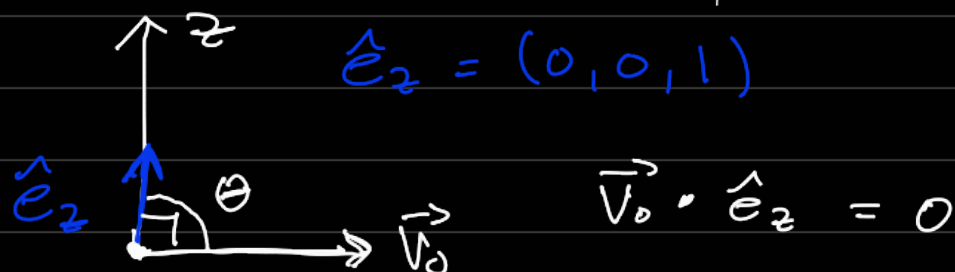
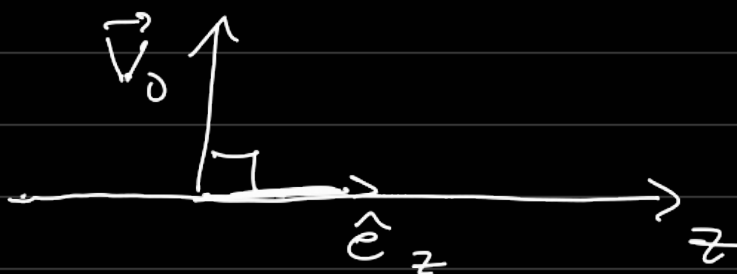
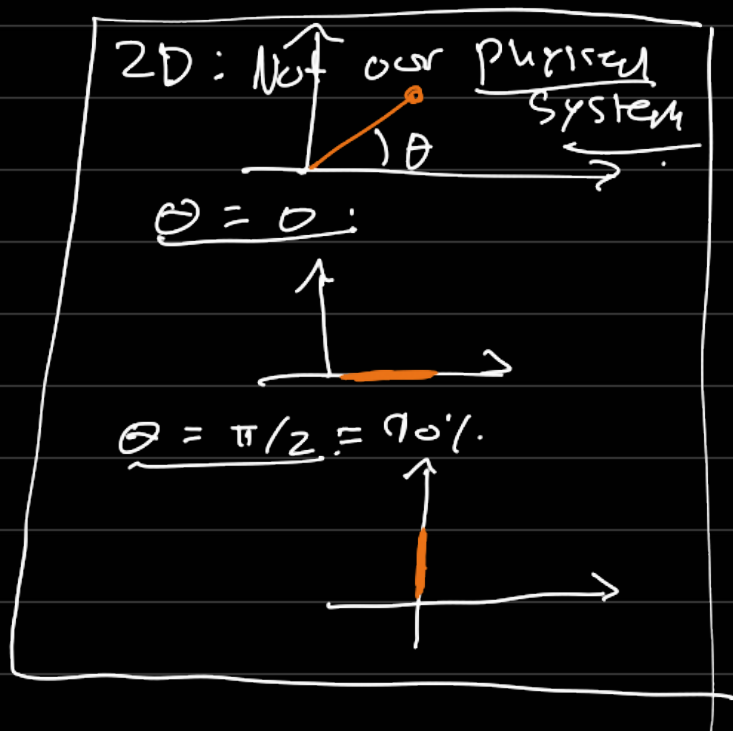
$$\bullet m \frac{d\vec{v}}{dt} = q \vec{v} \times \vec{B}$$

$$\textcircled{a} \vec{B} = B_0 \hat{e}_z, \textcircled{b} \vec{v}_0 \cdot \hat{e}_z = 0$$

$\hookrightarrow$ Plug this in for B (used $\textcircled{a}$)

$$m \frac{d\vec{v}}{dt} = q \vec{v} \times (B_0 \hat{e}_z) = B_0 q (\vec{v} \times \hat{e}_z)$$

Now, use $\textcircled{b}$: $\vec{v}_0 \cdot \hat{e}_z = 0 = |\vec{v}_0| |\hat{e}_z| \cos(\theta)$



$$\vec{v} = (v_x, v_y, 0)$$

$$v_z = 0$$

* Can use this: on both sides:

$$m \frac{d\vec{v}}{dt} = q B_0 (\vec{v} \times \hat{e}_z)$$

$$\hookrightarrow \begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ v_x & v_y & v_z \end{vmatrix} = v_y \hat{e}_x - v_x \hat{e}_y + 0 \hat{e}_z$$

$$\begin{vmatrix} 1 & 0 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} v_y & v_z \\ 0 & 1 \end{vmatrix} \hat{e}_z - \begin{vmatrix} v_x & v_z \\ 0 & 1 \end{vmatrix} \hat{e}_y + v_y \hat{e}_x - v_x \hat{e}_y$$

$$\vec{v} \times \hat{e}_z = v_y \hat{e}_x - v_x \hat{e}_y$$

LHS: $x \neq B_0$

$$\text{LHS: } m \frac{d\vec{v}}{dt} = m \frac{dv_x}{dt} \hat{e}_x + m \frac{dv_y}{dt} \hat{e}_y + m \frac{dv_z}{dt} \hat{e}_z$$

• Vector equations are "3-in-1"

• x-comp: are equal

y-comp: " "

z-comp: " "

X-comp: LHS: $m \frac{dv_x}{dt}$
 RHS: $q B_0 v_y$

$$m \frac{dv_x}{dt} = q B_0 v_y$$

$\vec{0} = (0, 0, 0)$

* Example: $\vec{A} + \vec{B} = \vec{0}$

$$\vec{0} = A_x \hat{e}_x + A_y \hat{e}_y + A_z \hat{e}_z + B_x \hat{e}_x + B_y \hat{e}_y + B_z \hat{e}_z$$

$$= (A_x + B_x) \hat{e}_x + (A_y + B_y) \hat{e}_y + (A_z + B_z) \hat{e}_z$$

* $\begin{cases} A_x + B_x = 0 \\ A_y + B_y = 0 \\ A_z + B_z = 0 \end{cases}$

$$\vec{A} = -\vec{B}$$

$$A_x \hat{e}_x + A_y \hat{e}_y + A_z \hat{e}_z = -B_x \hat{e}_x - B_y \hat{e}_y - B_z \hat{e}_z$$

* above my claim:

$$\begin{aligned} A_x &= -B_x \rightarrow \\ A_y &= -B_y \rightarrow \\ A_z &= -B_z \rightarrow \end{aligned} \begin{cases} A_x + B_x = 0 \\ A_y + B_y = 0 \\ A_z + B_z = 0 \end{cases}$$

$$m \frac{d\vec{v}}{dt} = m \frac{dv_x}{dt} \hat{e}_x + m \frac{dv_y}{dt} \hat{e}_y + m \frac{dv_z}{dt} \hat{e}_z$$

$$= q B_0 V_y \hat{e}_x - q B_0 V_x \hat{e}_y + 0 \hat{e}_z$$

$$x: m \frac{dV_x}{dt} = q B_0 V_y$$

$$y: m \frac{dV_y}{dt} = -q B_0 V_x$$

$$z: \frac{dV_z}{dt} = 0$$

Mixed ODEs:

Define Constant:

$$\omega \equiv \frac{q B_0}{m}$$

$$x: \frac{dV_x}{dt} = \omega V_y$$

$$y: \frac{dV_y}{dt} = -\omega V_x$$

$$z: \frac{dV_z}{dt} = 0$$

• Take the time deriv. of x:

$$\frac{d^2 V_x}{dt^2} = \omega \frac{dV_y}{dt}$$

$$\text{replace: } \frac{d^2 V_x}{dt^2} = \omega (-\omega V_x)$$

$$\frac{d^2 V_x}{dt^2} = -\omega^2 V_x$$

• Do the same for y:

$$\frac{d^2 V_y}{dt^2} = -\omega \frac{dV_x}{dt}$$

replace

$$\frac{d^2 V_y}{dt^2} = -\omega (\omega V_y)$$

$$\frac{d^2 V_y}{dt^2} = -\omega^2 V_y$$

Second order
ODE

$$\frac{d^2}{dt^2} = -\omega^2$$

* Complex functions: $e^{i\theta} = \cos(\theta) + i\sin(\theta)$

→

X-comp: $\frac{d}{dt} \left(\frac{dV_x}{dt} \right) = - \underbrace{\omega \cdot \omega}_{\omega^2} V_x$

$$\frac{dV_x}{dt} \sim \omega V_x$$

$$\frac{d^2 V_x}{dt^2} \sim \omega^2 V_x$$

• $\frac{d}{dt} e^{at} = a e^{at}$

$$f(t) = e^{at}$$

$$\frac{df}{dt} = a f$$

• $V_x = e^{i\omega t}$

$$\frac{d^2 V_x}{dt^2} = \omega^2 e^{i\omega t}$$

$$V_x = e^{-i\omega t} \rightarrow \frac{d^2 V_x}{dt^2} = (-\omega)^2 V_x = \omega^2 V_x$$

Real answer:

$$V_x = e^{i\omega t}$$

$$\frac{dV_x}{dt} = (i\omega) e^{i\omega t}$$

$$\frac{d^2 V_x}{dt^2} = (i\omega)^2 e^{i\omega t}$$

$$= i^2 \omega^2 e^{i\omega t}$$

$$= -\omega^2 e^{i\omega t}$$

• Note $e^{-i\omega t}$ is also an answer

$$(i\omega)^2 = (-i\omega)^2$$

• But $\underline{e^{i\omega t} + e^{-i\omega t}}$ is also an answer

"Most general" form

→ Comes from $e^{i\theta} = \cos(\theta) + i\sin(\theta)$

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

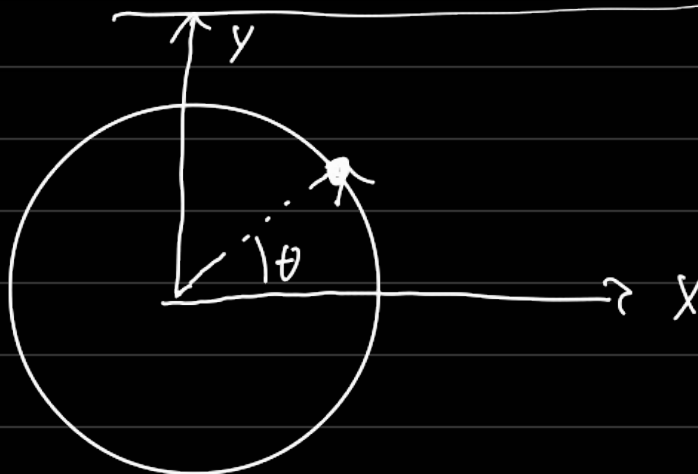
$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\sin(\theta) = \frac{1}{2i} (e^{i\theta} - e^{-i\theta})$$

Most general:

$$V_x, V_y = A \cos(\omega t) + B \sin(\omega t)$$

- Why it makes sense to have trig answer
 → Describing uniform circular motion:



$$V^2 = V_x^2 + V_y^2$$

$$\frac{dV_x}{dt} = \omega V_y \quad \text{and} \quad \frac{dV_y}{dt} = -\omega V_x$$

$$\frac{dV_x}{dt} = \omega V_y$$

V_x : cannot depend on $\cos(k)$

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\frac{d}{dx} \cos(x) = -\sin(x)$$

$$V_x = A \sin(\omega t)$$

*

$$\frac{dV_x}{dt} = A \frac{d}{dt} \sin(\omega t) = A \omega \cos(\omega t)$$

$$V_y = A \omega \cos(\omega t)$$

*

$$R_{ij} = \hat{e}_i \cdot \hat{e}_j$$

$$R_{xz} = \hat{e}_x \cdot \hat{e}_z \dots$$

